VLM Meets Latent Reasoning

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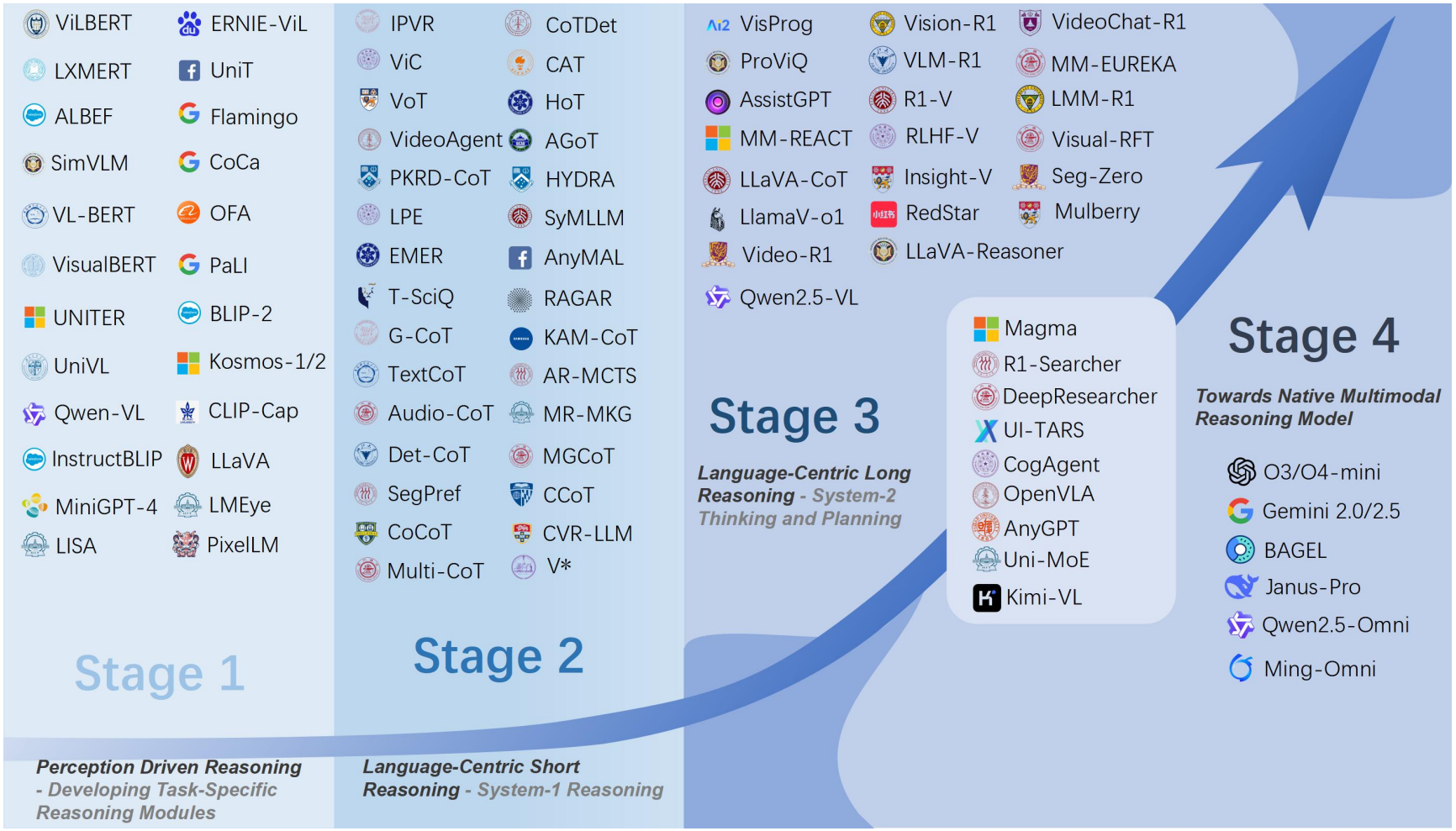
研究背景

- 近年得益于思维链技术的兴起，大型多模态/语言推理模型（MLRM/LRM）具备了显示思考、规划的能力，在各个任务上表现非凡。



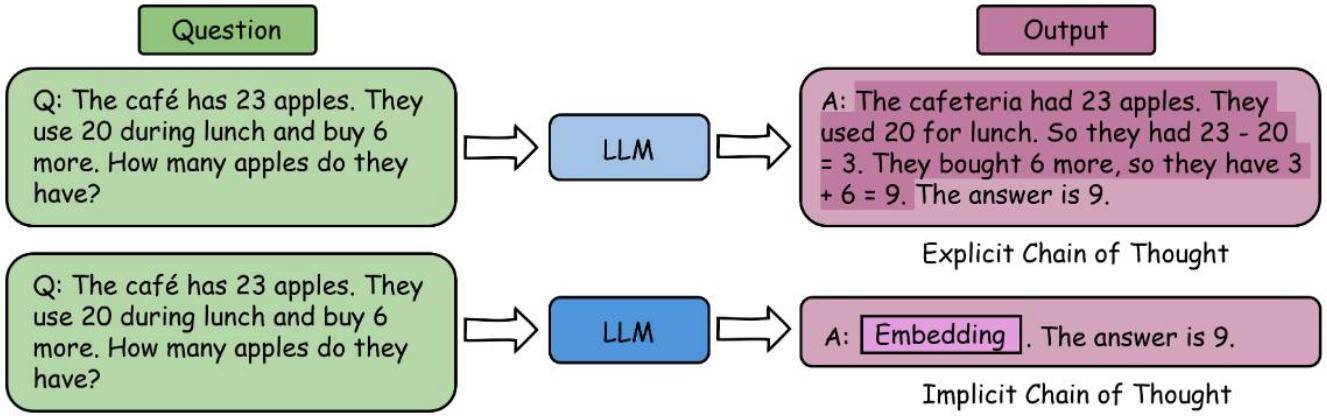
研究背景

显式COT的问题：

- 推理模型将推理过程必须以单词的形式展现出来，然而这与人的某些推理过程是违背的，比如设计图纸，空间推理，作曲等等。现有通过图像思考（Thinking with images）的方法依赖外部工具或显式辅助图像，灵活性受限，且难以实现类人抽象视觉思考。

- 每个特定令牌所需的推理量变化很大，然而当前的LLM架构为预测每个令牌分配了几乎相同的计算预算。推理链中的大多数记号仅仅是为了流畅性而产生的，很少涉及实际的推理过程。

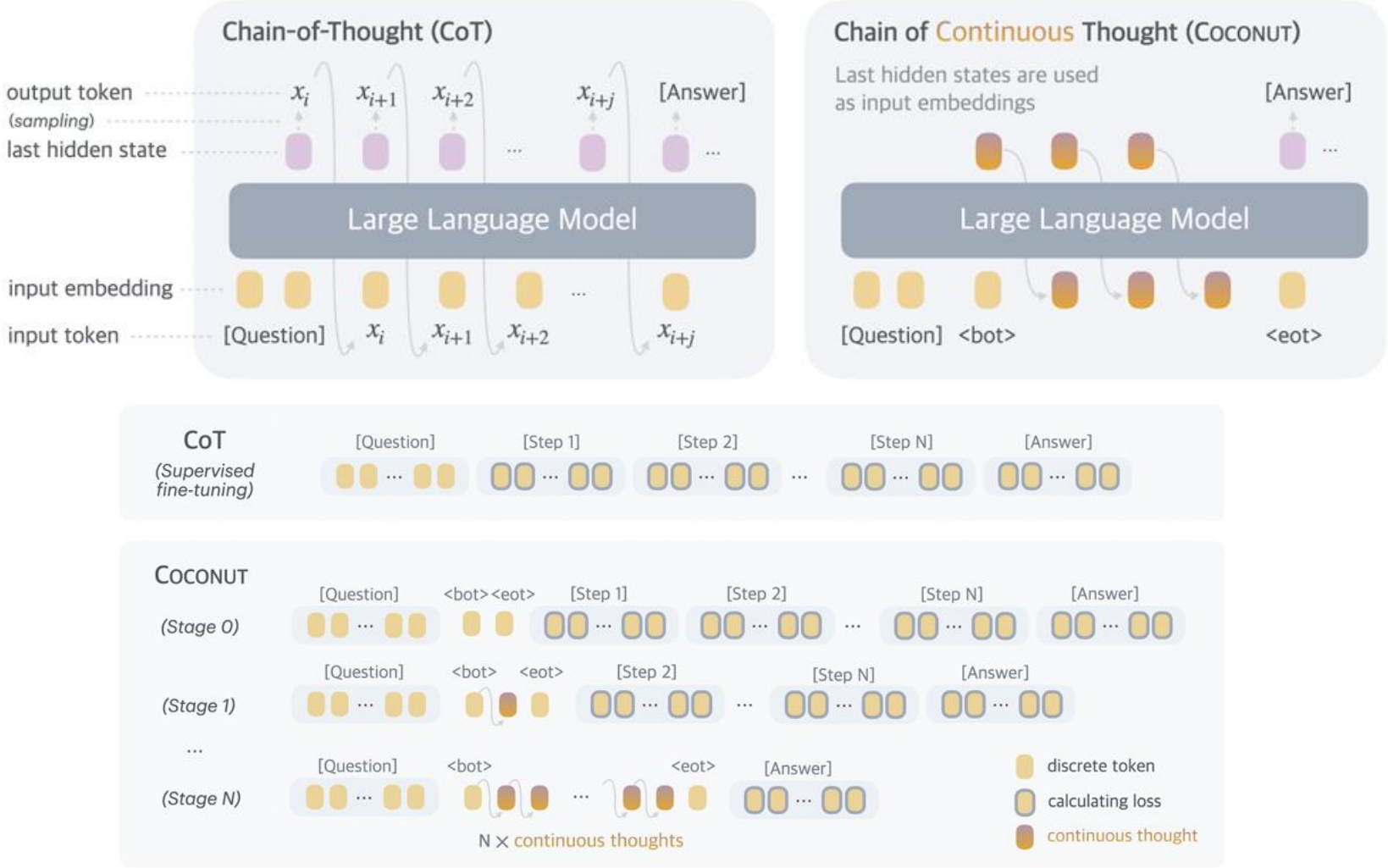
- 传统COT每生成一个token就做了不可逆决策，走错了只能硬编。



研究背景

- Hao, Shibo, et al. "Training large language models to reason in a continuous latent space, 2024." *Meta AI*.

把推理从语言空间搬到了hidden state，一个向量同时编码多条路径。



VLM Meets Latent Reasoning

- MLLMs 如何超越文本和外部工具，实现类人抽象视觉推理？

Monet: Reasoning in Latent Visual Space Beyond Images and Language，CVPR 2026

- 具身智能如何平衡长链推理与实时行动的效率冲突？

Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning，CVPR 2026

- 如何在不修改模型参数的前提下，提升多模态图像生成的推理能力和质量？

MILR: Improving Multimodal Image Generation via Test-Time Latent Reasoning，ICLR 2026

Monet

动机

- 现有 MLLMs 过于依赖文本 CoT 或外部工具进行视觉推理，效率低、不够灵活、难以捕捉细粒度视觉信息。

- 缺乏一种机制能让模型在内部以“视觉”的方式思考，而不是将视觉信息全部转化为文本。

Monet

核心思想:

允许多模态大语言模型在**连续潜在视觉空间**中直接进行推理，生成潜在视觉思考token。

- 一个思考过程中生成视觉token的例子，在COT的过程生成辅助图像Auxiliary image

Monet-SFT-125K Example 2: Creating New Visual States (from Zebra-CoT 3D Object Counting)

Question: Consider the initial arrangement of objects presented in the image below. You are required to perform a series of removal operations in the specified order:
Step 1: Eliminate all objects identified as PYRAMIDs.
Step 2: Eliminate all objects that are BLUE in color.
Step 3: Eliminate all objects that are RED in color.
After completing these steps, determine the total count of orange objects remaining.
Put your final answer within `\\boxed{}`."

Question Image:

Image-text Interleaved CoT:
I will begin by analyzing the initial set of objects shown in the problem image. There are `<observation>`7`</observation>` distinct objects in the initial configuration. My approach will be to systematically apply each removal step and visually track the remaining objects before arriving at the final count. The first operation requires the removal of all pyramid-shaped objects from the scene. This means both the `<observation>`purple pyramid`</observation>` and the `<observation>`orange pyramid`</observation>` will be taken away. To visualize the updated set of objects, I will generate an image reflecting this change.

After removing the pyramids, as seen in the previous visual, we are left with `<observation>`five`</observation>` objects: an `<observation>`orange sphere`</observation>`, a `<observation>`blue cube`</observation>`, a `<observation>`red torus`</observation>`, a `<observation>`red cube`</observation>`, and a `<observation>`purple sphere`</observation>`. The next instruction is to remove all objects that are blue. This specifically targets the `<observation>`blue cube`</observation>`. I will generate a new visual to show the state after this removal.

...

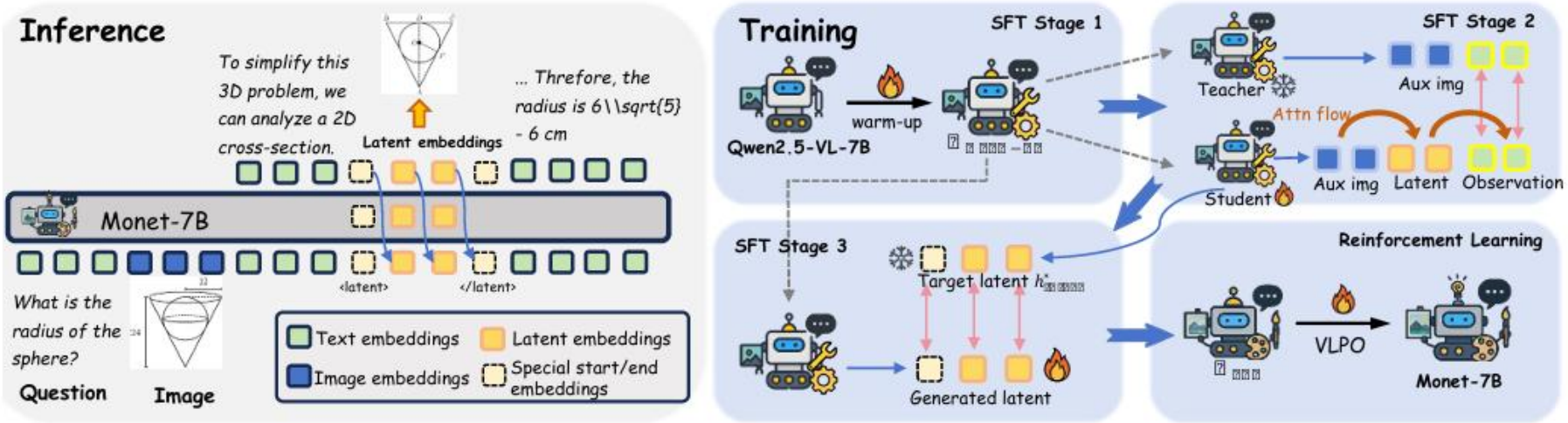
Having completed all specified removal operations, the final visual shows `<observation>`two`</observation>` objects remaining: an `<observation>`orange sphere`</observation>` and a `<observation>`purple sphere`</observation>`. The question asks for the count of orange objects. From the two remaining objects, only one is orange. The final answer is $\boxed{1}$.

Monet: Reasoning in Latent Visual Space Beyond Images and Language (CVPR 2026)

Monet

贡献点:

- 提出了**潜在视觉推理 (Latent Visual Reasoning)** 的新范式。
- 引入了**VLPO (Visual-latent Policy Optimization)** 强化学习方法优化潜在嵌入。
- 构建了高质量的**Monet-SFT-125K** 文本-图像交错 CoT 数据集。



Monet: Reasoning in Latent Visual Space Beyond Images and Language (CVPR 2026)

Monet

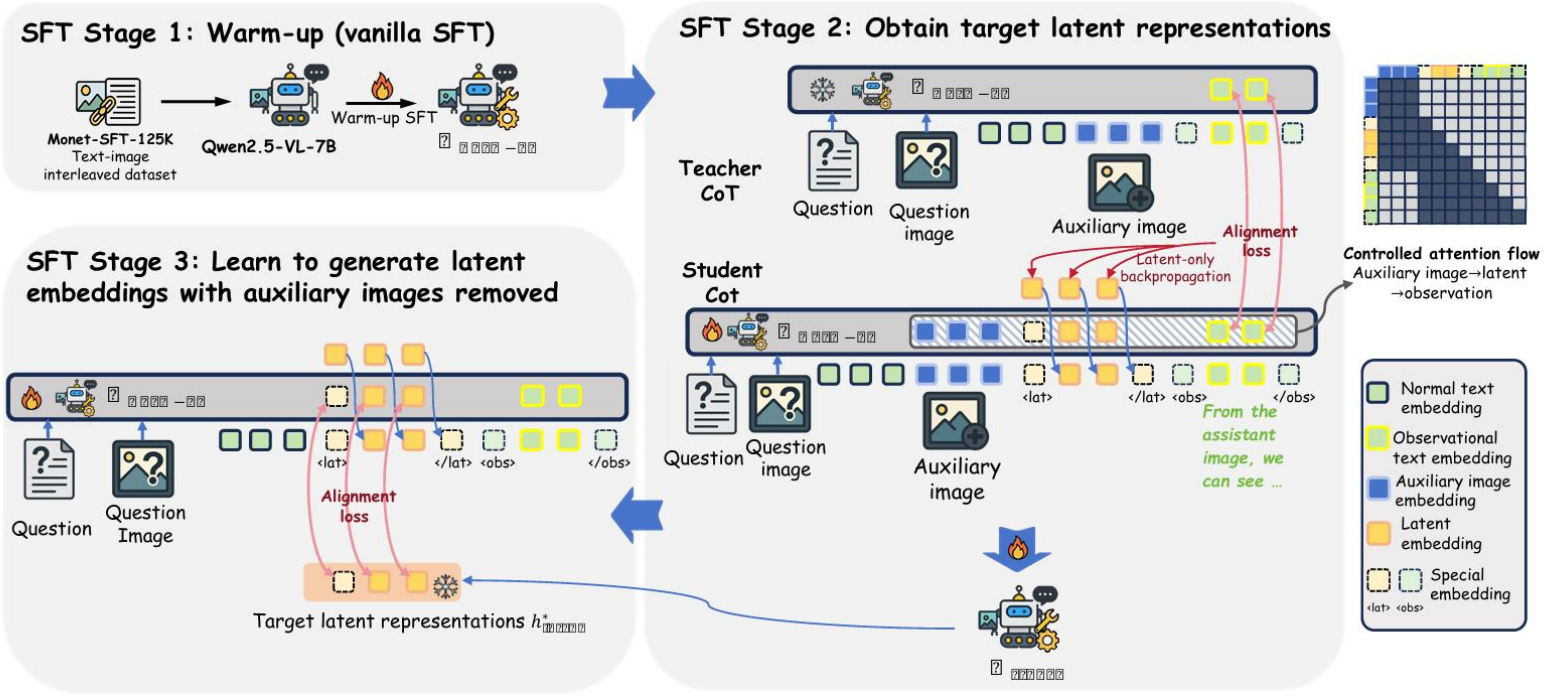
方法

三阶段蒸馏式监督微调框架:

阶段一: 视觉嵌入初始化: 利用预训练的视觉编码器将图像转化为潜在视觉嵌入。

阶段二: 潜在思考生成: 训练 MLLM生成与文本 CoT 相对应的连续潜在视觉嵌入，作为中间视觉思考。强调对潜在嵌入的梯度控制和双重监督（既有语言监督，也有潜在空间本身的监督）。

阶段三: 强化学习优化: 使用 VLPO 进一步优化潜在嵌入的生成策略，使其更能促进正确的最终回答。VLPO 作为 GRPO 的改进，专注于潜在推理



Monet: Reasoning in Latent Visual Space Beyond Images and Language (CVPR 2026)

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Model	V*			HRBench4K			HRBench8K			MME-RealWorld-Lite		
	Overall	Attribute	Spatial	Overall	FSP	FCP	Overall	FSP	FCP	Overall	Reasoning	Perception
Proprietary Model												
GPT-4o [16]	67.5*	72.2*	60.5*	59.0*	70.0*	48.0*	55.5*	62.0*	49.0*	52.0*	48.3*	54.4*
Open-Source Model												
Qwen2.5-VL-7B [1]	76.44	77.39	75.00	68.00	80.25	55.75	63.75	73.75	53.75	45.75	39.73	49.62
+ vanilla SFT	81.68	**83.48**	78.95	68.38	78.28	58.5	61.63	70.75	52.50	51.28	46.4	54.41
+ vanilla SFT + GRPO	78.53	78.26	78.95	70.00	83.25	56.75	66.75	78.00	55.50	52.42	48.13	55.18
Deepeyes [49]	**83.25**	**84.35**	81.58	**71.25**	83.75	**58.75**	65.13	77.00	53.25	54.28	50.53	56.63
LVR [20]	80.6*	81.7*	79.0*	-	-	-	-	-	-	-	-	-
Our Model												
Monet-7B (SFT + VLPO)	**83.25**	83.48	**82.89**	71.00	**85.25**	56.75	**68.00**	**79.75**	**56.25**	**55.50**	**51.07**	**58.34**
Relative Improvement	+6.81	+6.09	+7.89	+3.00	+5.00	+1.00	+4.25	+6.00	+2.50	+9.75	+11.34	+8.72

实验结果

- 在真实世界感知与推理任务 上表现出强大的泛化能力。
- 在抽象视觉推理任务上取得 SOTA 性能，验证了潜在视觉思考的有效性。

Monet: Reasoning in Latent Visual Space Beyond Images and Language (CVPR 2026)

Fast-ThinkAct By Nvidia

动机

- 具身智能和机器人操作任务需要实时、高效的决策。
- 传统 VLA 模型依赖生成冗长的文本 CoT (如 Thought-Action-Observation 链)，导致推理延迟高，无法满足实时性要求。
- CoT 的显式性也可能带来信息冗余，或在复杂场景下难以保持一致性。

Fast-ThinkAct

核心思想:

通过**可言语化的潜在规划 (Verbalizable Latent Planning)** 实现紧凑而高性能的 VLA (Vision-Language-Action) 任务推理，解决传统 CoT 的高延迟问题。

Question: Is this room on the ground floor of the building?
GT: no

Let's carefully analyze the image to determine if this room is on the ground floor of the building.

First, I notice that there is a window with a view of greenery outside. This suggests we are at a level that allows natural light to filter in, which is typical for ground floor windows.

Next, I see a white desk and a chair, which seem like standard office furniture for a ground floor workspace. The desk appears to be at a height that would be comfortable for someone to sit and work, which is also consistent with a ground floor layout.

Additionally, there's a white bookshelf that looks like it could be used for storage and decoration, which is a common feature in office spaces on the ground floor.

However, there is a cabinet in the background that seems a bit unusual for a ground floor room. It could be a storage cabinet for a specific purpose, but it might not be typical for a ground floor space.

Let's also consider the lighting. The natural light coming through the window seems to be coming from the side, which is more likely if we are on the ground floor and looking towards a window that is part of the building's exterior.

After considering all these factors, I think the room is likely on the ground floor, but it's not impossible for it to be on an upper floor with a unique window design or a specialized storage cabinet.

Final thought: Given the typical layout of office spaces and the visible elements, it's reasonable to conclude that this room is on the ground floor.

(a) Textual reasoning (from teacher model)

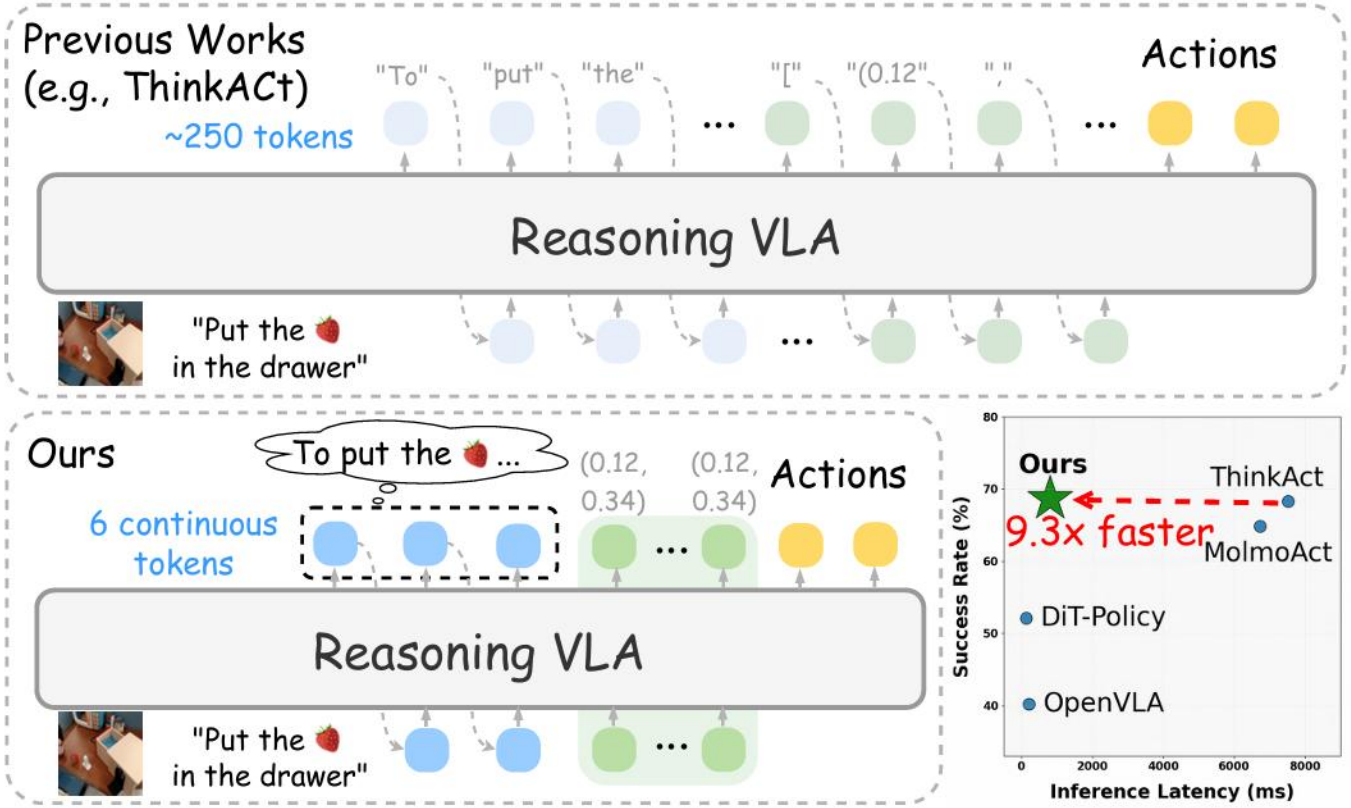
Let's break this down. The image shows a study room scene with a desk, a chair, and some shelves. There's no clear indication that this is on the ground floor. It's obviously indoors, but I should check if anything suggests a different level. Wait, I should also consider if there's any sign of elevation. Hmm, it seems like the view outside the window shows that the room is positioned above the ground. Oh, I see, the perspective through the window suggests it's not on the ground floor. Let's confirm that. Yes, it checks out.

(b) Verbalized reasoning

Fast-ThinkAct

贡献点:

- 引入**可言语化的潜在规划**概念，将高层规划压缩到紧凑的潜在表示中。
- 提出通过**偏好引导的蒸馏** 学习高效潜在规划。
- 实现了显著**降低推理延迟**，同时保持或提升了具身任务性能。



Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning， CVPR 2026

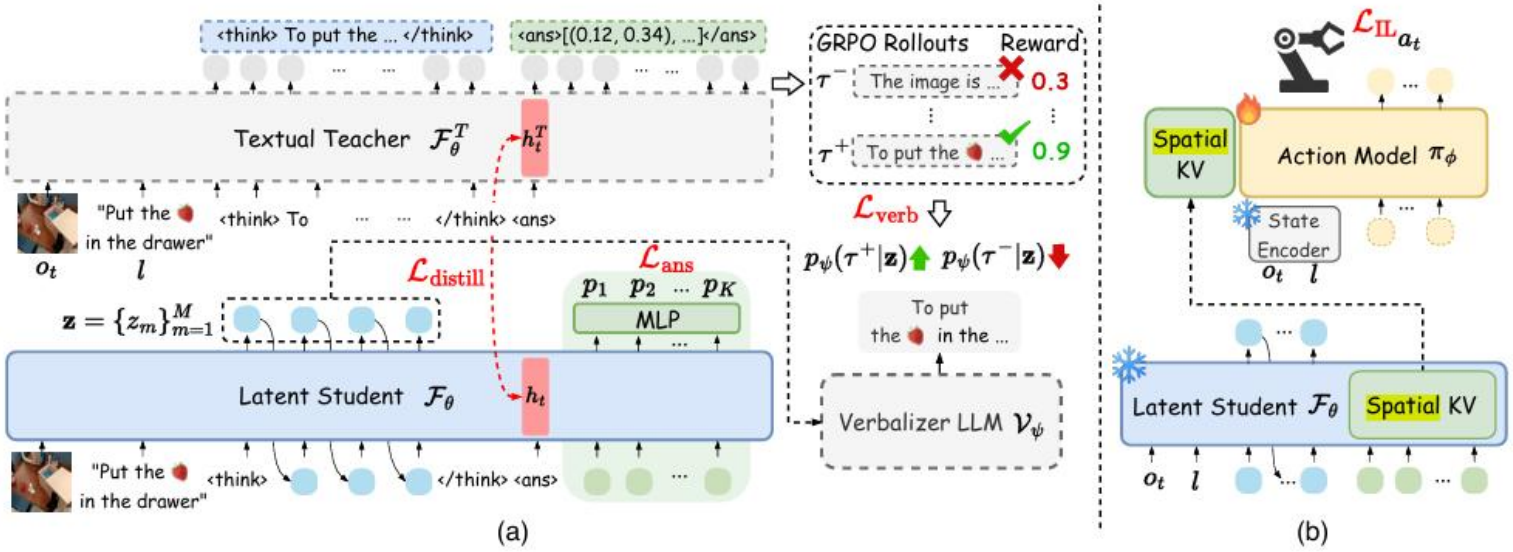
Fast-ThinkAct

方法:

教师模型 (Teacher Model): 一个传统的、基于 CoT 的 VLA 模型 (如 ThinkAct)，生成冗长的文本推理过程和相应的操作轨迹。

学生模型 (Student Model - Fast-ThinkAct):

- **潜在规划模块:** 从教师模型中蒸馏，学习生成紧凑的、连续的潜在规划，这些潜在规划可以被解码回文本以供解释。
- **偏好引导的**通过对齐教师模型的操作轨迹和潜在规划所驱动的操作轨迹，将规划能力从教师转移到学生。这确保了潜在规划不仅紧凑，而且有效。
- **推理增强的策略学习** 将潜在规划作为高层信息输入到低层动作策略中，实现从抽象规划到具体行动的无缝衔接。



Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning， CVPR 2026

Fast-ThinkAct

Method	LIBERO	SimplerEnv-Google	Latency (↓)
OpenVLA-7B Kim et al. (2024)	76.5	40.2	N/A
CoT-VLA-7B Zhao et al. (2025)	83.9	N/A	N/A
ThinkAct-7B Huang et al. (2025)	84.4	68.3	7513
MolmoAct-7B Lee et al. (2025)	86.8	64.9	6723
ThinkAct-3B Huang et al. (2025)	83.1	64.7	5674
Fast-ThinkAct-3B	**89.7**	**68.7**	**805 (↓7.0×)**

Model	click alarm		click bell		turn switch		adjust bottle		beat block		handover mic		handover block		hanging mug		stack blocks two		stack bowls three		Average	
	E	H	E	H	E	H	E	H	E	H	E	H	E	H	E	H	E	H	E	H	E	H
DP Chi et al. (2023)	61	5	54	0	36	1	97	0	42	0	53	0	10	0	8	0	7	0	63	0	43.1	0.6
ACT Zhao et al. (2023)	32	4	58	3	5	2	97	23	56	3	85	0	42	0	7	0	25	0	48	0	45.5	3.5
π_0 Black et al. (2024)	63	11	44	3	27	23	90	56	43	21	98	13	45	8	11	3	42	1	66	24	52.9	16.3
RDT Liu et al. (2024)	61	12	80	9	35	15	81	75	77	37	90	31	45	14	23	16	21	2	51	17	56.4	22.8
ThinkAct Huang et al. (2025)	64	13	84	11	40	19	94	70	79	33	92	40	56	15	31	18	30	5	54	23	62.4	24.7
Fast-ThinkAct	70	17	82	12	37	21	92	72	82	33	99	42	65	15	30	22	45	5	55	25	**65.7**	**26.4**

实验

- 在多样化的具身操作和推理基准测试中取得了强大的性能。
- **推理延迟大幅降低，**显著提升了具身智能体的实时响应能力。

Fast-ThinkAct: Efficient Vision-Language-Action Reasoning via Verbalizable Latent Planning， CVPR 2026

MILR

动机

- 现有图像生成模型 (如扩散模型) 在生成复杂、知识密集型图像时，推理能力不足，可能产生不一致或不准确的内容。
- 传统的微调方法需要大量数据且修改模型参数，不灵活。
- 在生成一次图像后，很难修改或“纠正”生成过程中的潜在缺陷。

核心思想:

通过在**统一的潜在向量空间**中联合推理图像和文本，改进多模态图像生成。

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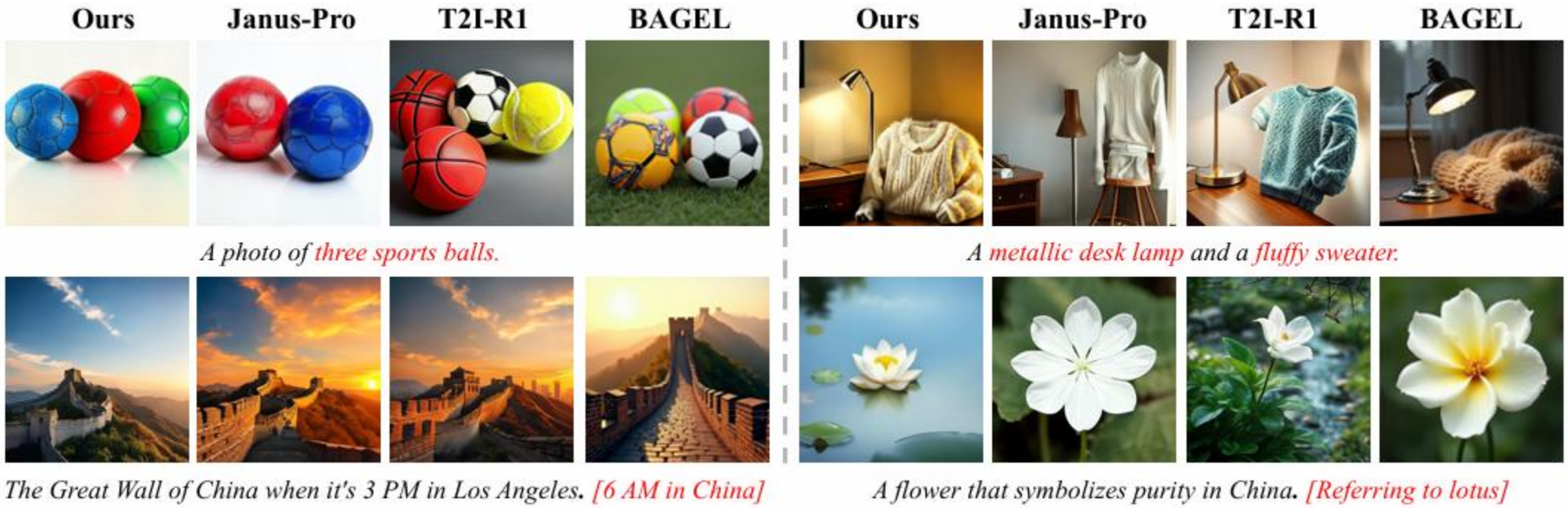
MILR: Improving Multimodal Image Generation via Test-Time Latent Reasoning (ICLR 2026)

MILR

贡献点:

- 提出一种**test-time**的多模态潜在推理方法，不修改模型参数。
- 在统一的潜在向量空间中，**联合优化图像和文本 token** 的表示。
- 利用**图像质量评估器**作为强化学习的奖励信号。



MILR: Improving Multimodal Image Generation via Test-Time Latent Reasoning (ICLR 2026)

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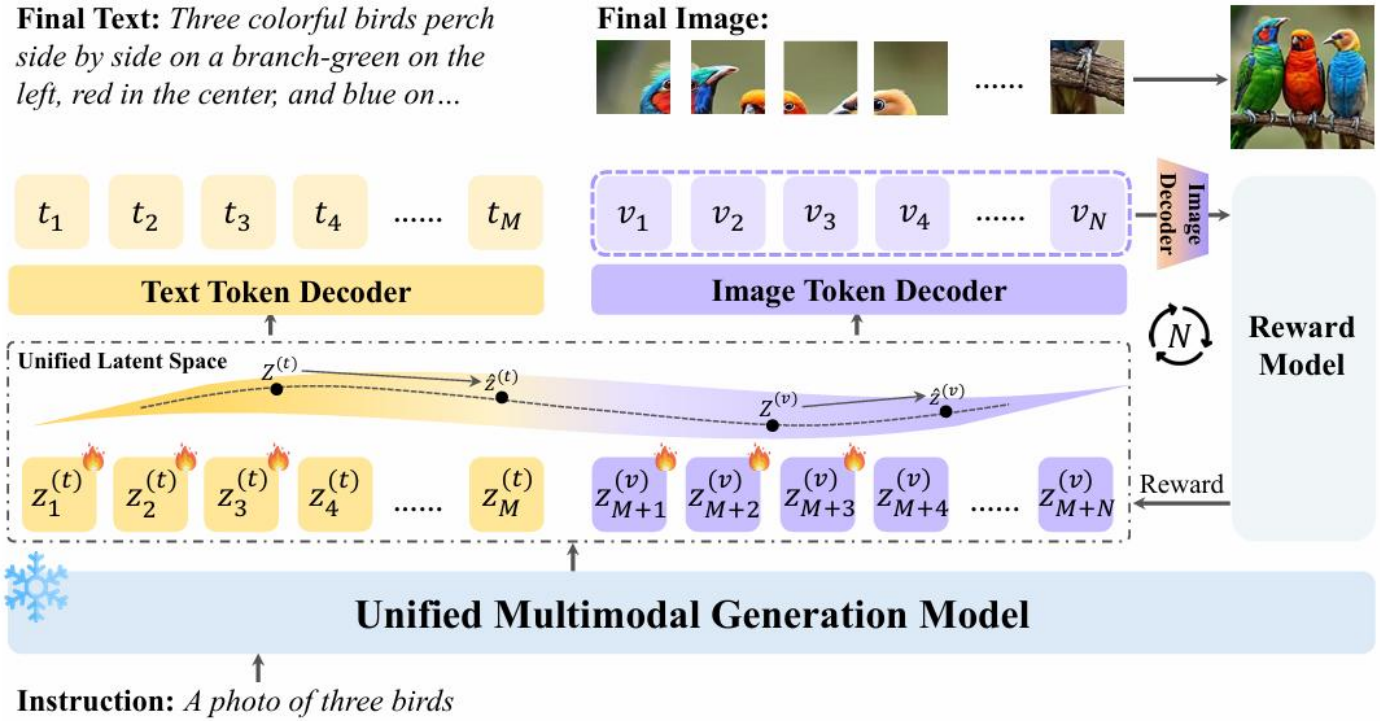
方法:

- **初始化:** 给定文本提示和初始图像 (可为空或粗略生成)。
- **潜在表示优化:** 通过策略梯度 迭代地优化文本 token 和图像 token 在统一潜在空间中的向量表示。
- **优化比例:** 文本 token 和图像 token 的优化比例可以调整，以适应不同任务需求。
- **奖励机制:** 使用**图像质量评估器** 对生成的图像进行评分，作为策略梯度更新的奖励信号。这使得模型能在推理时自我评估和改进。
- **迭代 refinement:** 重复优化过程，直到达到预设的迭代次数或收敛



MILR: Improving Multimodal Image Generation via Test-Time Latent Reasoning (ICLR 2026)

MILR

实验

- 在多个图像生成benchmark上取得 SOTA 结果。
- 展示了在复杂指令遵循、概念组合和细节生成方面的显著改进。

Method	Single Obj. ↑	Two Obj. ↑	Counting ↑	Colors ↑	Position ↑	Attr. Binding ↑	Overall ↑
Non-reasoning Models							
LlamaGen (Sun et al., 2024)	0.71	0.34	0.21	0.58	0.07	0.04	0.32
Emu3 (Wang et al., 2024)	0.98	0.71	0.34	0.81	0.17	0.21	0.54
FLUX.1-dev (Black Forest Labs, 2024)	0.98	0.79	0.73	0.77	0.22	0.45	0.66
DALL-E 3 (Betker et al., 2023)	0.96	0.87	0.47	0.83	0.43	0.45	0.67
SD3-Medium (Esser et al., 2024)	0.99	0.94	0.72	0.89	0.33	0.60	0.74
BAGEL (Deng et al., 2025)	0.99	0.94	0.81	0.88	0.64	0.63	0.82
GPT-4o (OpenAI, 2025)	0.99	0.92	0.85	0.91	0.75	0.66	0.85
Training-based Reasoning Models							
GoT-R1 (Duan et al., 2025)	0.99	0.94	0.50	0.90	0.46	0.68	0.75
T2I-R1 (Jiang et al., 2025a)	0.99	0.91	0.53	0.91	0.76	0.65	0.79
Flow-GRPO (Liu et al., 2025a)	**1.00**	**0.99**	**0.95**	0.92	**0.99**	0.86	**0.95**
ReasonGen-R1 (Zhang et al., 2025b)	0.99	0.94	0.62	0.90	0.84	0.84	0.86
Janus-Pro-7B(+GRPO) (Tong et al., 2025)	0.99	0.87	0.61	0.87	0.82	0.68	0.81
Janus-Pro-7B(+DPO) (Guo et al., 2025)	0.99	0.89	0.65	0.92	0.82	0.72	0.83
Test-time Reasoning Models							
Reflect-DiT (Li et al., 2025b)	0.98	0.96	0.80	0.88	0.66	0.60	0.81
ReflectionFlow (Zhuo et al., 2025)	**1.00**	0.98	0.90	0.96	0.93	0.72	0.91
Janus-Pro-7B(+Text Enhanced Reasoning)	0.98	0.91	0.55	0.89	0.74	0.67	0.79
Janus-Pro-7B(+Best-of-N)	0.99	0.96	0.89	0.93	0.92	0.80	0.91
Janus-Pro-7B(+PARM) (Guo et al., 2025)	**1.00**	0.95	0.80	0.93	0.91	0.85	0.91
Janus-Pro-1B (Chen et al., 2025)	0.98	0.82	0.51	0.89	0.65	0.56	0.73
Janus-Pro-1B+MILR	**1.00**	0.91	0.78	0.92	0.86	0.86	0.89
Janus-Pro 7B (Chen et al., 2025)	0.98	0.85	0.56	0.89	0.77	0.64	0.78
Janus-Pro-7B+MILR	**1.00**	0.96	0.90	**0.98**	0.98	**0.91**	**0.95**

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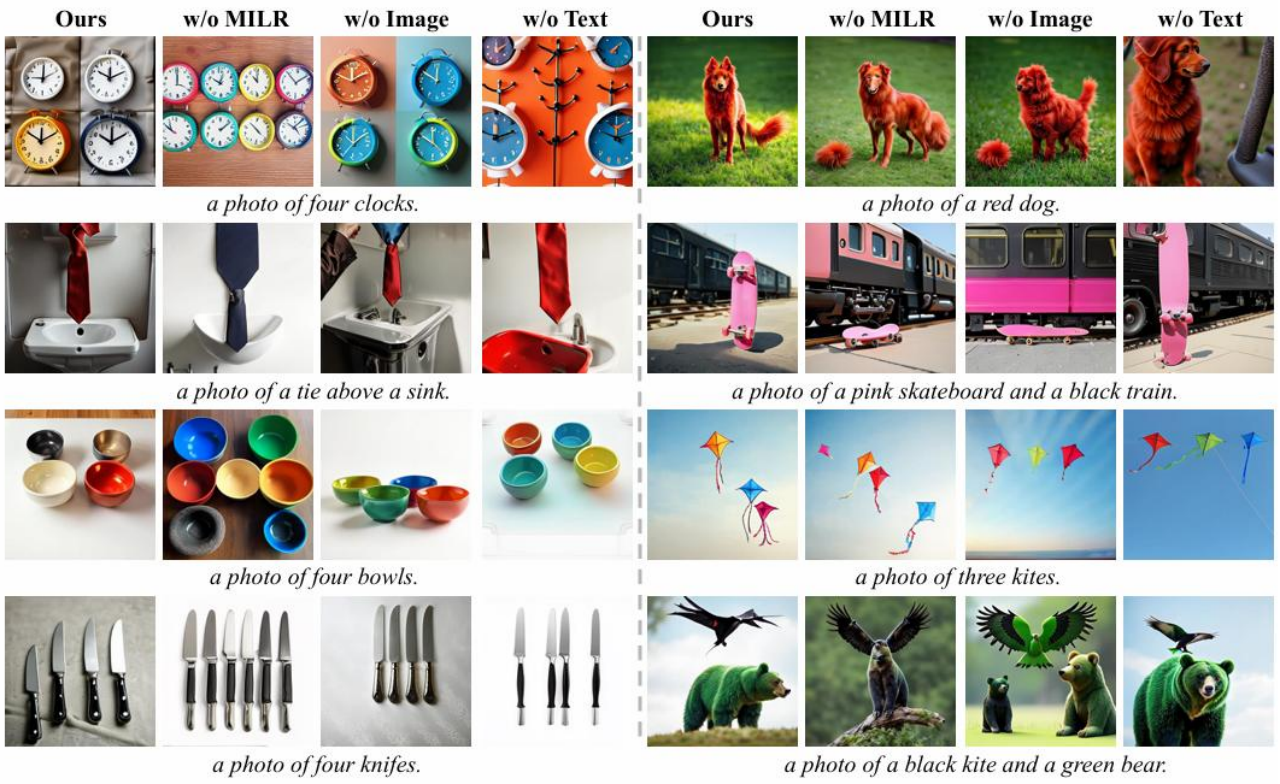
实验

- 组件的消融实验展示了对于每一个模态的latent优化都可以显著提升图像生成的质量



Rebuttal

- “推理”一词在这里存在争议。该过程更准确地说是一种引导式潜空间搜索，通过优化潜在向量以最大化奖励，而非结构化、逻辑化的思维过程（如大型语言模型中的推理定义）。

我们完全理解你的观点。在大型语言模型（LLM）的语境中，“推理”通常指的是明确的、逐步的逻辑推理，比如思维链（Chain-of-Thought，CoT）。然而，在本研究中，我们遵循了新兴的潜性推理文献，其中“推理”并不局限于自然语言中的显式CoT，而是指对潜态的迭代更新。例如，COCONUT [1] 明确定义推理为连续潜在空间中隐藏状态的重复更新，且不发出中间文本符号。用潜在推理扩展测试时计算[2]同样描述了一种循环深度架构，通过在测试时反复对隐藏表示应用块来“推理”，而不是产生更长的CoT痕迹。
MILR遵循了潜在推理的概念。如第3节所述，我们保持所有模型参数固定，并通过奖励引导的策略梯度目标迭代更新统一的多模态潜在表示，然后将优化的潜在状态解码为文本和图像标记。这种在潜空间中由反馈驱动的优化，而非显式符号的CoT，正是我们论文中所称的“推理”。

总结

三篇论文的共通之处:

- **核心: 潜在表示推理:** 都强调在连续的、紧凑的潜在空间中进行推理，以克服离散符号推理的局限性。
- **解决效率/泛化问题:** 致力于提高各自领域模型的推理效率和泛化能力。
- **利用强化学习策略:** 通过策略梯度或偏好引导的蒸馏来优化推理过程或潜在表示。
- **超越传统 CoT:** 都试图减少或优化对冗长文本 CoT 的依赖。

•**三篇论文的主要差异**:

•**应用领域**:

- M□∅◯⬀⬀: MLLM Ⓢ 的抽象视觉思考
- F∠Ⓢ⬀⬀–T↗≡∅⌒A▱⬀⬀: 具身智能/VLA
- MILR: 图片生成质量和指令遵循

•**潜在表示的粒度**:

- M□∅◯⬀⬀: 连续潜在**视觉嵌入**
- F∠Ⓢ⬀⬀–T↗≡∅⌒A▱⬀⬀: 可言语化的**潜在规划**
- MILR: 统一的**图像–文本潜在向量**

思考与启发

- 如何应用于各自的下游任务：

1.该任务有推理延迟要求

2.该任务的中间推理过程依赖文本无法精准体现推理过程

3.该任务中间推理过程的某些token对最终answer起到不可逆的致错作用

- 如何设计更有效的潜空间奖励和优化机制，潜空间的可解释性

请各位批评指正！